

# NANDINI SAXENA

Pune, Maharashtra, India | nandinisaxenawork@gmail.com | [Linkedin](#) | [Github](#) | [Medium](#)

## EDUCATION

### Bharati Vidyapeeth (DU), College of Engineering, Pune

Jul 2023 – May 2027

BTech, Computer Science & Engineering — CGPA 9.3/10

Relevant coursework: Machine Learning, Deep Learning, Natural Language Processing, Agentic AI, Business Intelligence, Cloud Computing

## EXPERIENCE

### Machine Learning Intern — DIAT, DRDO Pune

Jan 2026 – Apr 2026

- Built a transformer-based sequence model for UAV trajectory reconstruction during GPS outages using IMU sensor data, reducing trajectory RMSE by ~28% versus classical dead-reckoning integration
- Designed the dead-reckoning pipeline end-to-end — noise filtering, bias correction, and drift modelling for inertial-only navigation under GPS denial

### Vice President — Google Developer Groups (GDG) BVP Pune

Jun 2025 – Present

- Lead a 12-person core team running technical programming for a 1500+ member developer community
- Secured and manage 5+ industry partnerships for events, sponsorships, and speaker engagements

### Software Development Engineer Intern — MCarbon Tech

Jun 2025 – Jul 2025

- Built a smart ticketing system on Spring Boot, Flask, and Vue.js, covering backend services, API integration, and frontend delivery

## SELECTED PROJECTS (GITHUB.COM/NANDINI1612)

### [GateIO — Yaw-Rate-Gated Causal TCN for UAV GPS-Outage Bridging](#)

- Extended DRDO research: gated a causal temporal convolutional network with yaw-rate signal to bridge GPS outages in UAV navigation

### Disentangling Logical Validity from Statistical Utility in Synthetic Tabular Data

- Built a constraint-repair extension for hierarchical tabular generation models that enforces logical/business constraints on generated records while preserving the statistical utility of the original output

### [Phenological Mismatch Observatory](#)

- Combined iNaturalist and eBird observation data to compute phenology mismatch scores, validated to 6.7-day / 2.5-day MAE against ground truth; now expanding into a CV classifier for citizen-science photos

### [Chicago Crime Analytics Platform](#)

- Built a full-stack spatial-temporal analytics platform (Flask, React, PostGIS) processing 50,000+ real records from the Chicago Data Portal, with SMA-based crime forecasting

### [AI Financial Crime Detector](#)

- Compared GNN and XGBoost fraud detection under concept drift on the Elliptic Bitcoin dataset, quantifying performance degradation as transaction patterns shift over time

Additional: Fashion Colour vs. Economic Trend Analysis (VAR forecasting) · Conversational Analytics (CSAT prediction, SHAP explainability) · Neonatal EEG Seizure Detection (in progress)

## TECHNICAL WRITING

- Published explainers on Medium (@nandinisaxenawork): Prophet's seasonality decomposition, GNNs vs. simpler models under distribution drift, CNN inductive biases, and Markov chains

## SKILLS

**Languages:** Python, Java, SQL, JavaScript, C, TypeScript

**ML / DL:** PyTorch, PyTorch Geometric, TensorFlow, HuggingFace Transformers, scikit-learn, XGBoost, Statsmodels, SHAP, XGBoost

**Focus areas:** Time-series forecasting, Graph Neural Networks, Sequence Modeling, Concept-drift Analysis, Computer Vision

**Backend / Infra:** FastAPI, Flask, Spring Boot, React, PostgreSQL, Git, AWS, GCP

## LEADERSHIP & ACTIVITIES

- Founder, OpporYouNity (youth opportunity & mental-health platform) · Head of PR, Null Student Chapter · Classical Bharatnatyam dancer, 10+ years (RAQS College Dance Team)